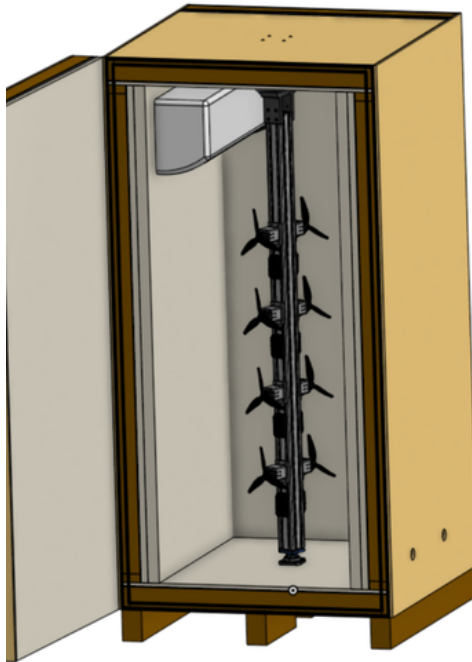




Motor Life Test Chamber - Internship

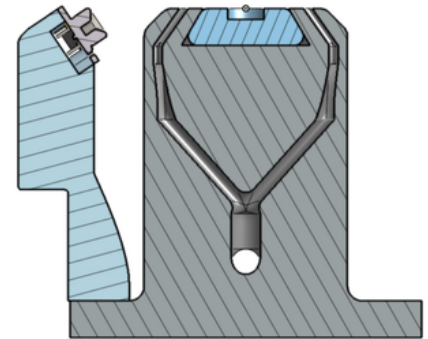


Overview

The Motor Life Test Chamber runs motors through long or short endurance tests using different throttle blip profiles, and measures efficiency by reading current and voltage at the ESC. It runs 8 motors at once, and an actuator up top rotates them to mimic the gyroscopic loading a motor sees in flight. I've had a hand in many improvements to the design and process, but below are my biggest contributions.

Active Motor Cooling

I designed the cooling manifolds directly into the motors' 3D-printed mounts, fed by a single fitting into each mount. Routing the air through the mount itself put cooling right at the motor and cut the time to cool in half.

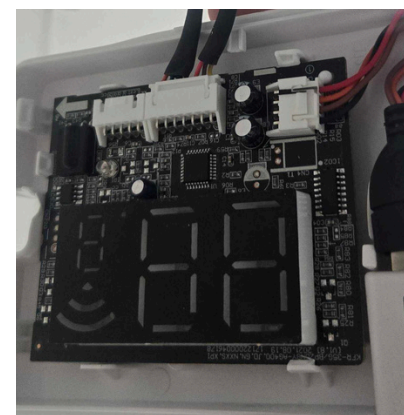


Power Supply Box

I designed the internal layout and built all the wire harnesses: 9 power supplies at roughly 15 kW total, 9 power harnesses, and a CAN bus to each actuator and motor. I laid it out to fit the supplies as compactly as possible while still leaving full access to every terminal for wiring and service. It has never failed. No shorts, no wiring issues, and no downtime across more than 13,000 official 3-minute throttle blip cycles, and it still runs fine.

Temperature Control (PID AC)

I put the chamber's air conditioning under closed-loop PID control by hacking its IR receiver. I harvested the AC's IR command signals and retransmitted them from an Arduino and an IR transmitter, so the controller could drive the unit directly, allowing for complete automated cycling. I tuned the loop myself and built in hysteresis so the AC compressor cycles gently instead of short-cycling, which protects its lifespan while still holding a stable test temperature.





Motor Life Test Chamber - Internship



Operator Process Improvements

Before, there was barely any desk space to assemble motors and no defined test cycle, so swapping motors in and out was slow and clunky. I mapped the workflow with a flow chart, wrote an SOP, and reorganized the space around it. Custom jigs now handle assembling the dovetails on the motors, and each motor has a set place so it's clear where it goes when it's boxed for shipping. The result is a clean, repeatable testing process. Overall cycle time dropped 10 min to 4 min from all of my improvements.

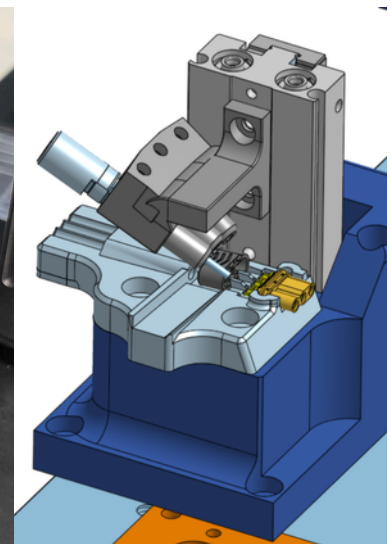
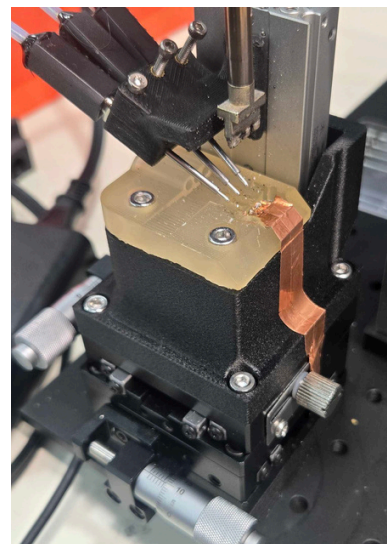
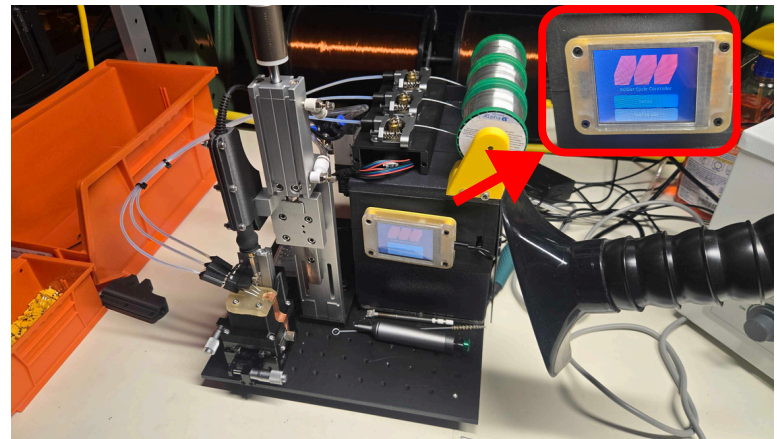
Custom MR30 Wire Harness Soldering Machine - Internship

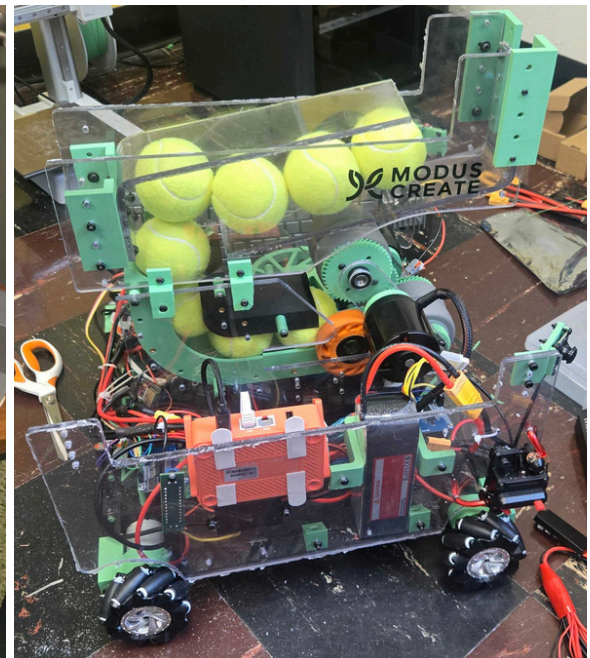
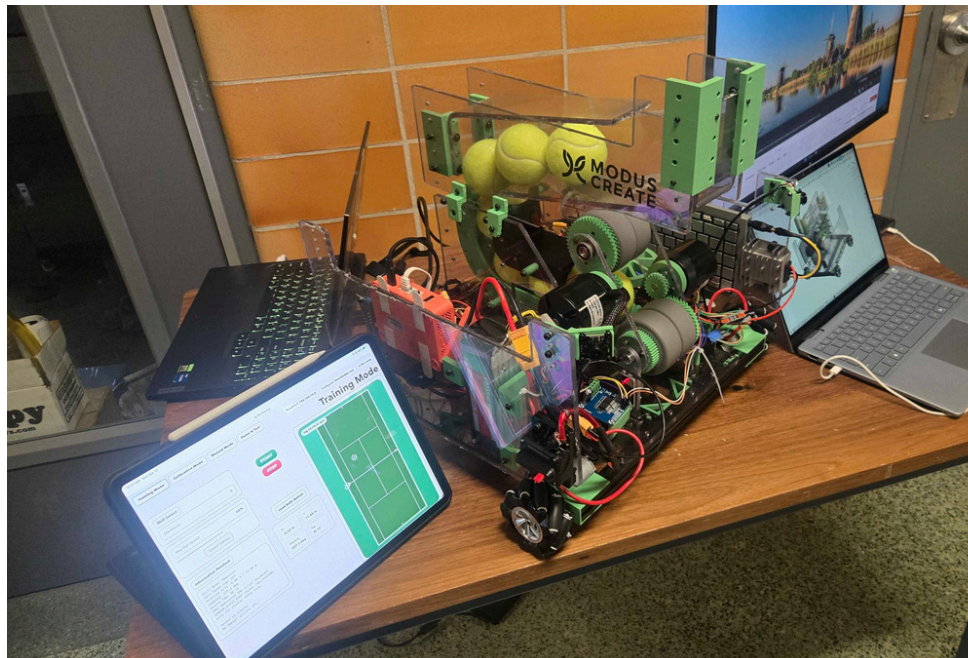
Overview & Design

A soldering machine we ordered from a Chinese supplier ran over a 70% scrap rate on MR30 wire harnesses, far too high for production. I built our own machine around the parts that worked and cut scrap to 10-20% at one harness every 12 seconds. I reverse-engineered the original tip-down mechanism and used Onshape parametric part studios to design fixtures that reuse its hardware and align the connector. An X-Y optics table lets us fine-tune the assembly to absorb part tolerances and misalignment down to 0.01mm. Two pneumatic actuators run it, one for the iron tip and one for the solder needles. I had a working prototype in a weekend.

Controls & My Role

The machine runs off an ESP32 Cheap Yellow Display board, where the operator sets the soldering profile: temperature, dwell, needle timing, and extrusion length. I did all the mechanical design and built the fixtures. Another person handled most of the software, but I set up SSH and made some changes myself, including better calibration and flashing stall protection and CoolStep onto the stepper driver board.





Tennis Buddy is an autonomous-capable robot that acts as a tennis training partner, firing consistent volleys with adjustable spin. It has a full drivetrain, a flywheel shooter, an actuated hood, gravity-fed loading, onboard cameras, and a laptop interface. We designed and built all of it in 24 hours, including machining the polycarbonate plates.

My Role

I led the robot's design and built the shooting and hood mechanisms from concept to finished hardware. I also helped finalize the drivetrain, loading, and control subsystems so the full system came together inside the hackathon window.

Shooting Mechanism

Two 4000 W BLDC skateboard motors spin the flywheels. Running the two wheels at different speeds sets the spin, so the robot can put topspin or backspin on the ball or hit it flat.

Hood Mechanism

The hood is servo-actuated through herringbone gears, which keep the motion smooth and cut backlash. It fits in a tight envelope and adjusts the launch angle repeatably from shot to shot. This is the subsystem I'm most proud of.

Ball Handling

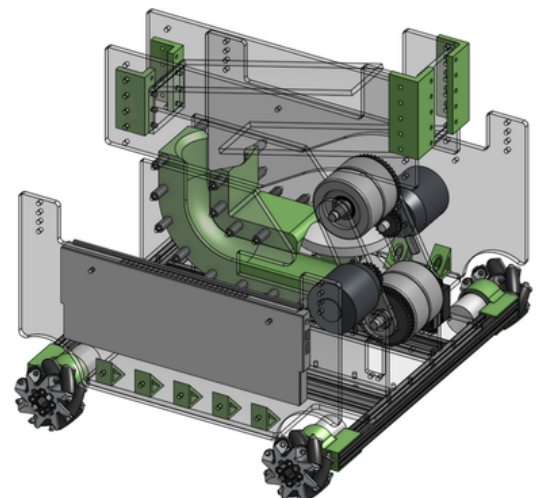
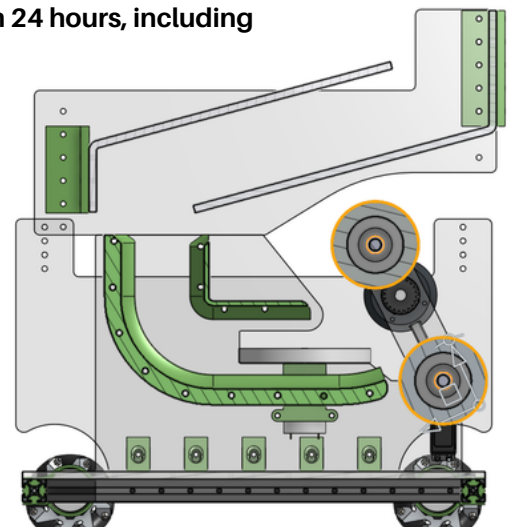
The robot holds 9 balls and feeds them by gravity. A compliant-wheel indexer separates one ball at a time and loads it into the shooter for continuous volleys.

Drivetrain

Mecanum wheels give the robot omnidirectional movement, so it can reposition and hold placement anywhere on the court.

Control & Vision

An ESP32 handles Wi-Fi control, and the program run

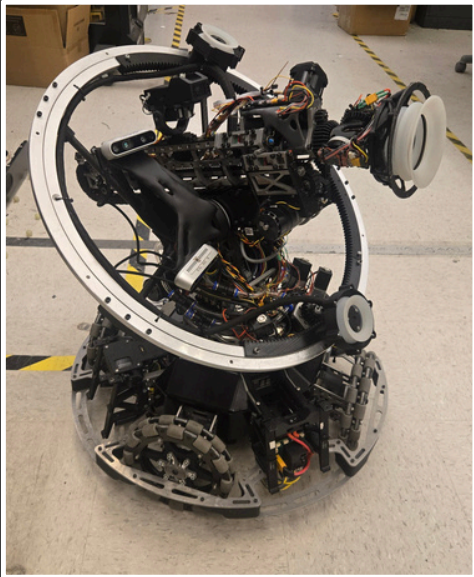
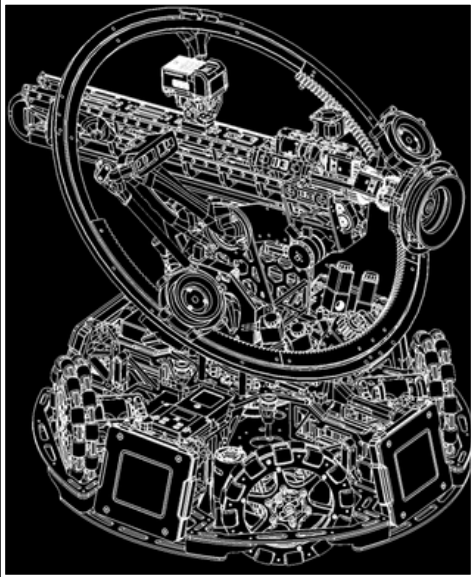


As one of six team members, I contributed to the design and construction of a 0.6 × 0.6 m, 20 lb robot built for a competitive challenge requiring precise cube manipulation and placement. This project demanded careful consideration of mechanical tolerances, hardware selection, and control systems to achieve high accuracy.

Key Features:

- 5 rotational axis enabling versatile manipulation
- Cascading arm design with ~1 m reach
- Electrical and pneumatic slip rings on turret yaw and wrist roll for continuous rotation
- Three passive encoded omni-wheels providing precise odometry
- Suction-based end effector for reliable object handling

KRONOS



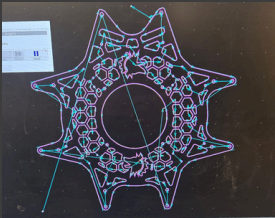
My work

Full Manufacturing life-cycle experience with plate and cylidrical geometries

Manufacturing

Design

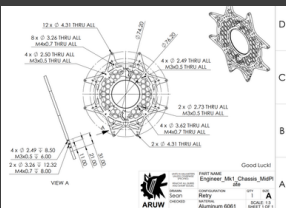
Chassis Midplate



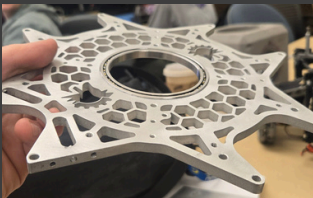
Waterjet operation



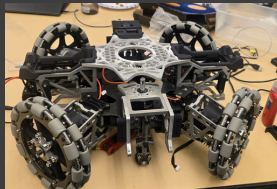
surface post-processing



Part drawings using GD&T principles



Machining and tapping with fixture plate



Assembly

Turret bearing stack plates

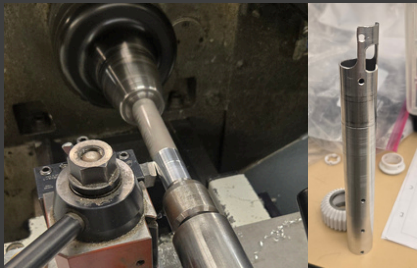


2-axis conversational milling

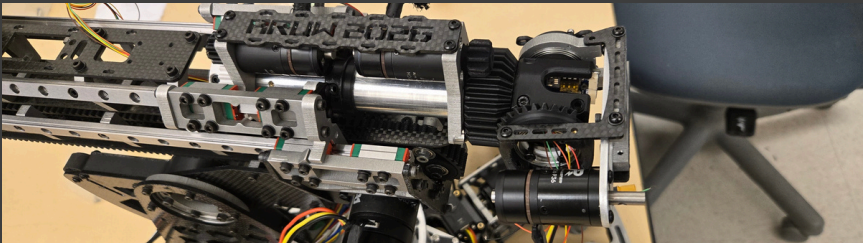


precise bearing press-fits

Wrist Tube



Part requiring lathe and mill operations



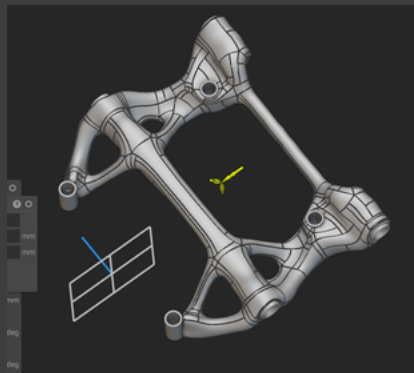
Full assembly of my designed wrist mechanism

My work

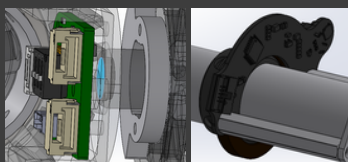
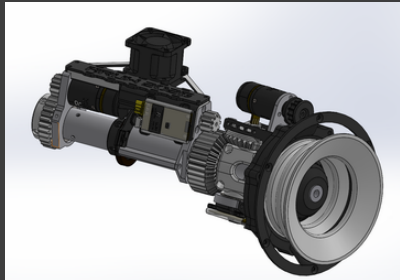
Main Project:

2-DOF Wrist Mechanism (Primary Contribution)

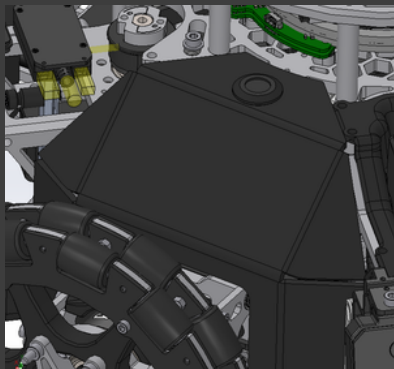
- Continuous 360° roll via slip-ring integration
- 180° pitch → hemispherical orientation workspace
- Low backlash, high stiffness design through gear iteration
- On-axis encoding for precise control



Topology optimized electronics mount (nTop)



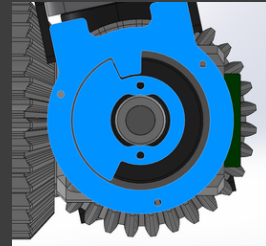
On-axis encoding on 2-axis



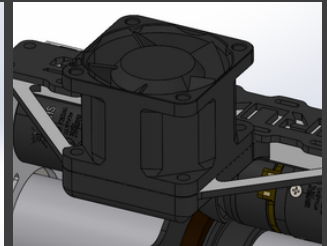
Electronics Covers

Manufacturing

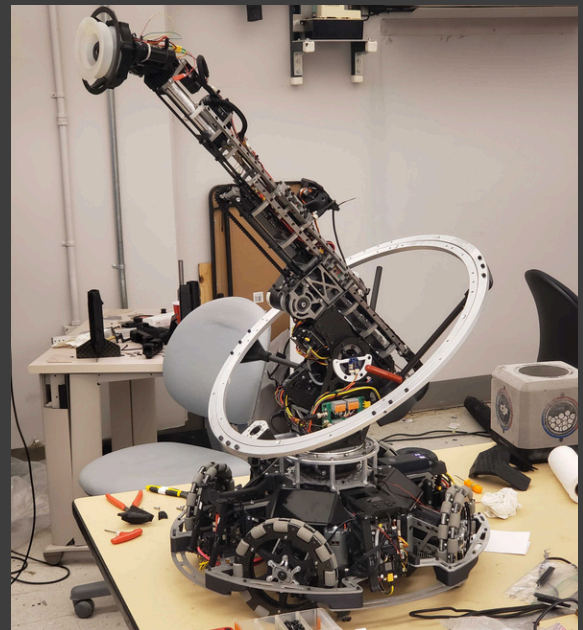
Design



Hardstops for safety and fallback while testing

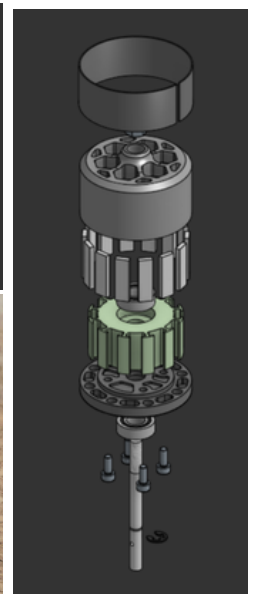
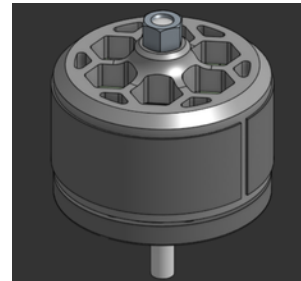


Active cooling for stall condition

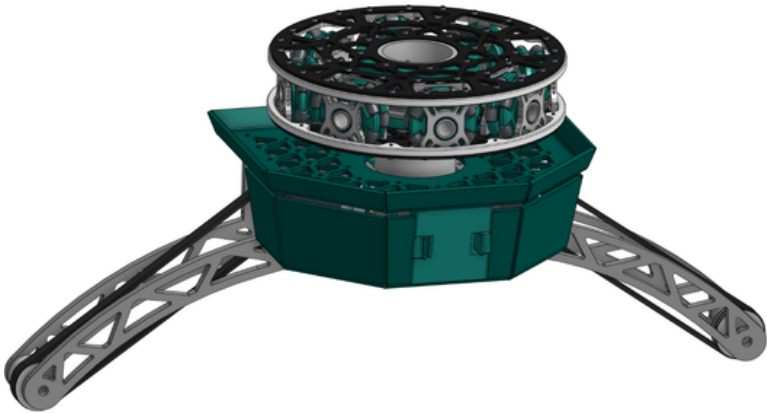


BLDC Motor Project - Learning Project (In progress)

- Designed a fully 3D printed BLDC motor
- Implemented flux loss solution from plastic stator by adding steel ring
- Balanced turn count vs slot fill to hit ~150 Kv at 24V

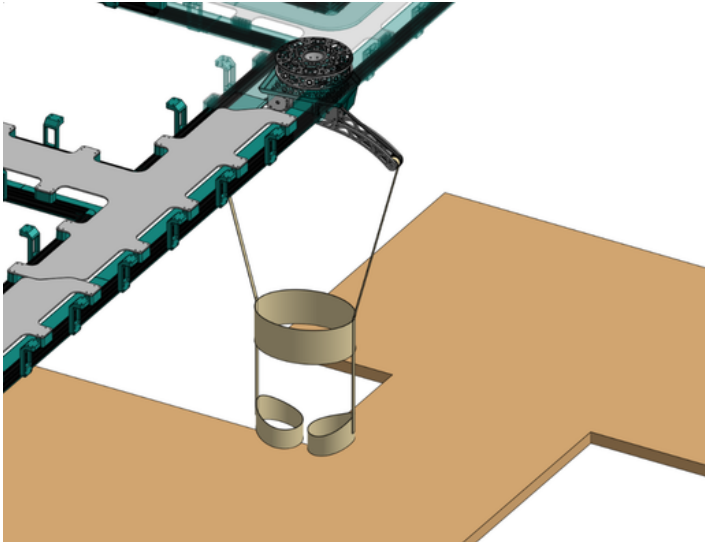


In a team of two, we set out to create a CAD project to solve a problem under the theme of "Wearable Technology"

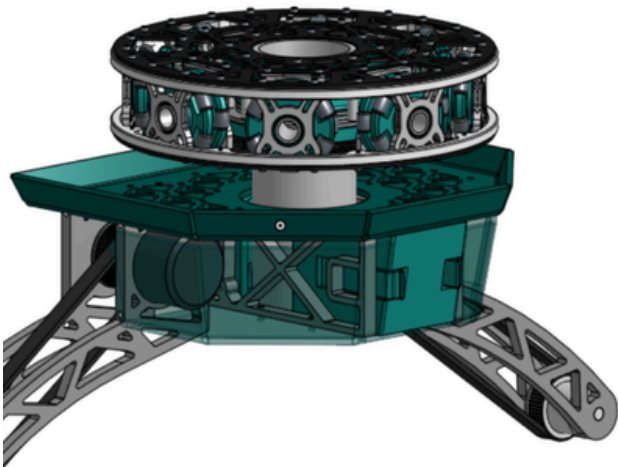


Carrier Assembly

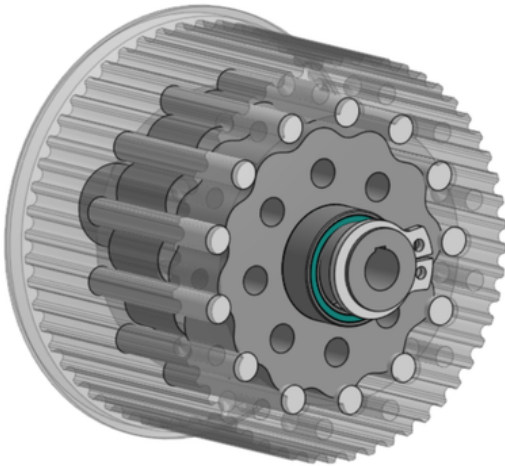
Our project is a robotic harness that can detect sudden torque changes and correct for them, saving individuals from falls, while giving them the freedom to move around.



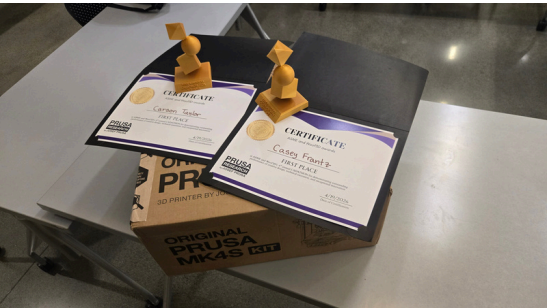
Demo room-arrangement, showing ceiling mounted configurable rails



Transparent view showing finger-accessible clip design, rotary mecanum wheel assembly, and motor placement



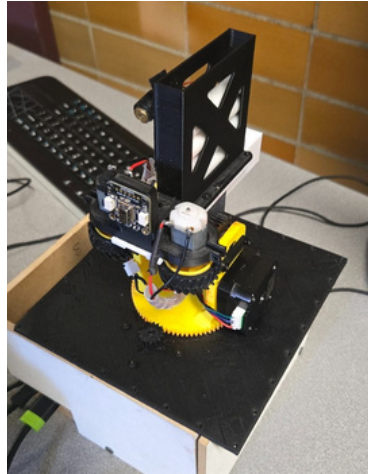
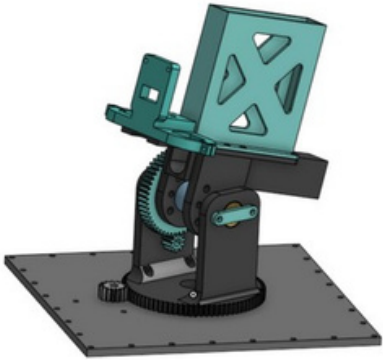
Dual-offset cycloidal gearbox integrated into a belt pulley for an ultra-compact seamless motor gearing



First Place Win



2 DOF Nerf Turret- Hardware Hackathon



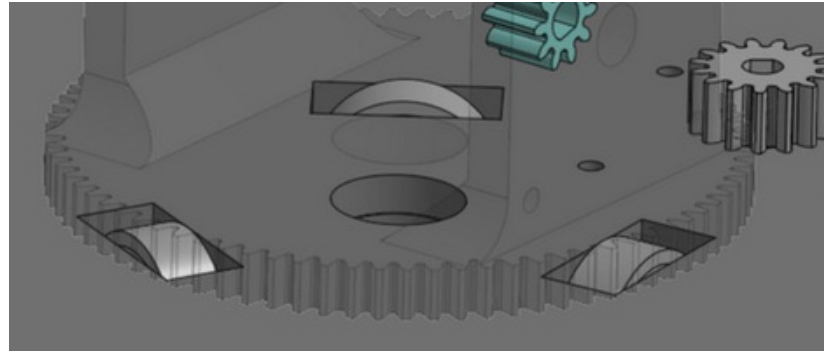
Overview:

A two-degree-of-freedom Nerf turret capable of 360° motion in the XY plane, featuring a gravity-fed magazine and custom thrust bearing for smooth rotation.

The system was controlled through a combination of motors, servos, an Arduino, and a Raspberry Pi.

Design Process:

The turret was designed, 3D printed, and assembled within 24 hours during a hackathon. I created multiple indexer designs for the servo-driven rotation system to improve rigidity, precision, and tolerance control. Rapid iteration and modular design allowed for quick troubleshooting and optimization during assembly.



Result:

My team won 2nd place out of 40 teams with high scores in complexity and completeness. We made multiple design considerations for the next iteration of this project such as improved motor and mounting for the shooting mechanism



Technical Highlights:

Tools: SolidWorks, Arduino IDE, Raspberry Pi

Skills: 3D printing, servo control, CAD modeling, mechanical integration

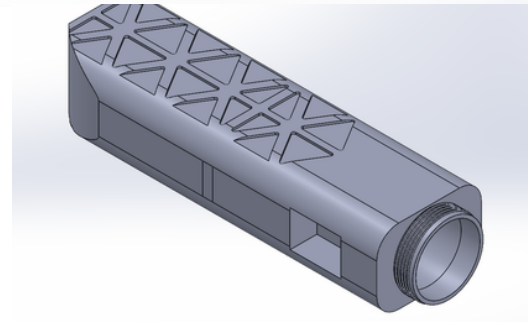
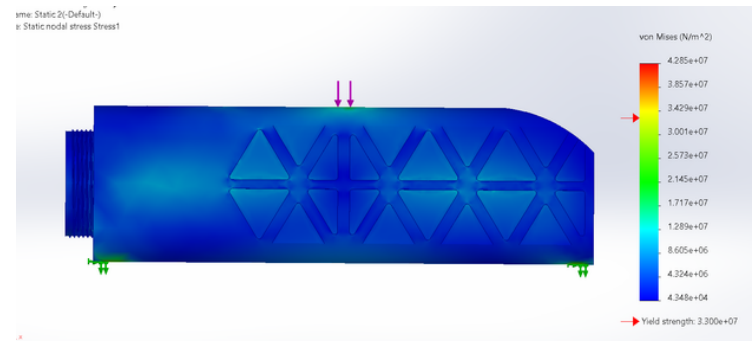
Methods: Rapid prototyping, tolerance optimization, embedded system control

Overview:

Collaborated with a team to design and optimize a lightweight flashlight body capable of withstanding a three-point bend test. The objective was to minimize deflection under load while maintaining a low overall mass.

Design Process:

Using SolidWorks Simulation (FEA), I performed design studies to evaluate different cross-sectional profiles, wall thicknesses, and material distributions. Through iterative refinement, I identified and reinforced high-stress regions while reducing unnecessary mass. The optimized geometry achieved the 3rd lowest deflection-to-weight ratio among all competing designs, demonstrating exceptional stiffness efficiency and strong performance relative to weight.



Tools: SolidWorks Simulation, Design Studies
Skills: Finite Element Analysis (FEA), lightweight design, structural optimization, stress evaluation
Methods: Three-point bend testing, iterative simulation, performance benchmarking

Beetle Weight Battlebot - Crimson Robotics

What?

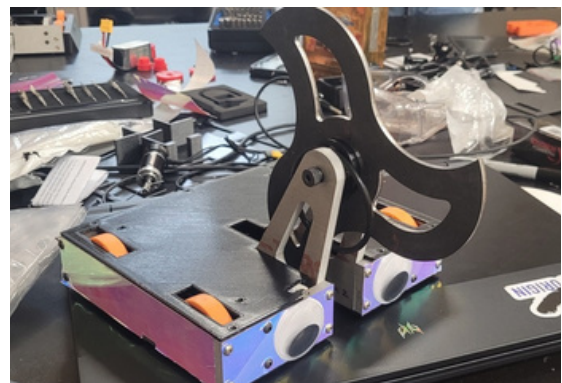
3 lb battlebot which was designed, built, and programmed all within limited time over 2 months. I have developed most parts of the CAD model which includes drive and weapon functionality.

How?

Robot concept was first drafted using Onshape which then was converted over to Solidworks. We looked over 3 iterations of drive base and a weapon design before arriving upon our final design choice.

Result.

Robot was entirely functional but had issues particularly with interference of motors and electronics along with fragility of motor mount design. This experience was a great learning opportunity.





Swerve Drive - Personal Design Project

Design Process:

Modeled in SolidWorks using advanced features such as wraps, sheet metal, weldments, and surface modeling. I analyzed component interfaces, refined clearances, and developed an exploded animation to illustrate motion and assembly. The project was completed with manufacturability and ease of assembly in mind.

Overview:

Designed and modeled a fully functional, manufacturable swerve drive assembly based on West Coast Products' Swerve X module. The drive system allows omnidirectional movement and incorporates precision tolerances suitable for real-world fabrication.



Results:

The final assembly included 43 unique parts with zero interferences and accurate mechanical motion. The swerve module demonstrated realistic operation and precise steering articulation across all degrees of freedom.



Technical Highlights:

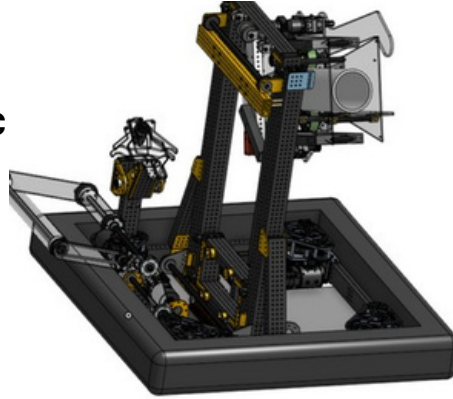
Tools: SolidWorks (Surface, Sheet Metal, and Motion Study)

Skills: CAD modeling, tolerance analysis, reverse engineering, animation rendering

Methods: Parametric design, interference validation, manufacturability optimization

Overview:

Served as a design mentor for a high school FIRST Robotics team during the FRC 2025 season. The robot featured a dual-stage cascading elevator, multi-object intake, and climbing mechanism for endgame scoring.

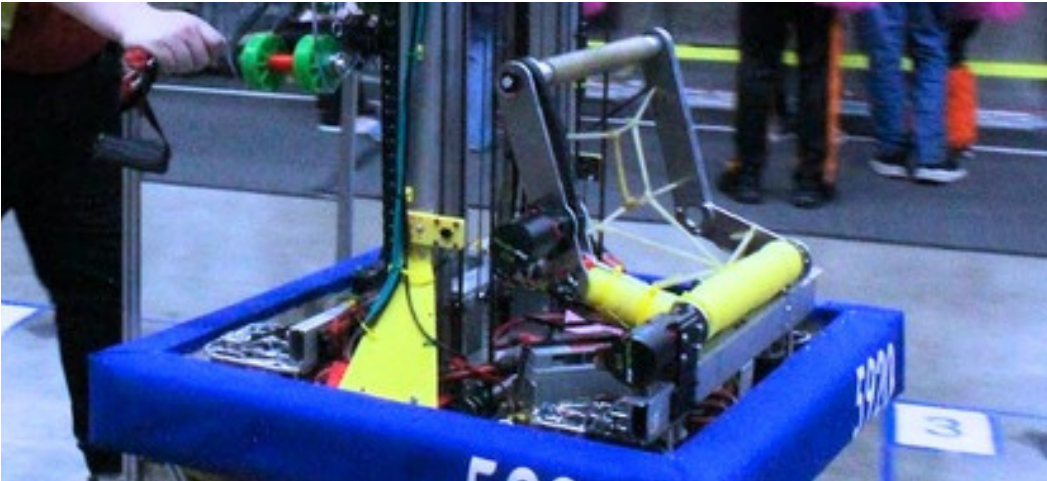


Design Process:

My focus was mentoring students through the CAD and design process for the intake and climb mechanisms. I introduced the team to CAD fundamentals, tolerance planning, and iterative prototyping. Together, we refined the slap-down intake for smoother actuation and improved game piece reliability.

Results:

The team overcame early mechanical issues and became one of the top five scorers in the Pacific Northwest regional competitions. The final robot was praised for its consistency, scoring efficiency, and durability across multiple matches.



Technical Highlights:

Tools: Onshape, SolidWorks

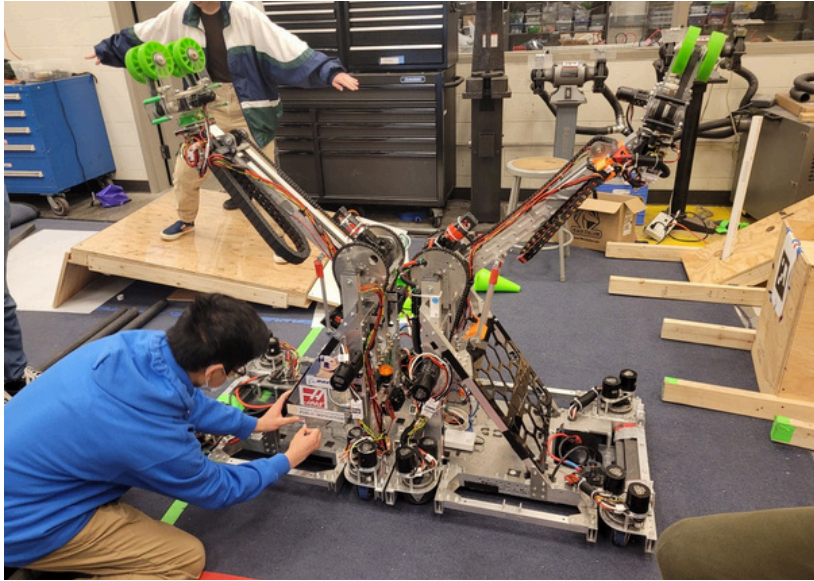
Skills: Mentorship, mechanical design, intake optimization, design for reliability

Methods: Iterative CAD refinement, student training, system integration

CASEY FRANTZ

FUTURE MECHANICAL ENGINEER STUDYING AT THE UNIVERSITY OF WASHINGTON

FIRST Robotics Competition - Mechanical Lead



What?

Two fully functional competition robots for [FRC's 2023 game](#). Both have a swerve drive base with a telescoping arm and two action intake which can pick up both cubes and cones securely to complete the game's objective

How?

Designed by our groups design subteams, it was sent off to me to lead the manufacturing of all aluminum, polycarbonate, and 3D printed parts. All parts were either manually milled, turned, or CNC routed with our machines.

Result.

Our robots performed outstandingly, leading us to the world championships, winning us engineering awards, and becoming top 40 of 3000 teams in the world in regard to placement. This was possible from fantastic integration of design, hardware, electronics, and software. Particularly, my impact was driving the robot and ensuring that the robot was entirely robust and reliable throughout multiple [high-contact competitions](#).

